

BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY
DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING

EEE416: MICROPROCESSOR AND EMBEDDED SYSTEM LABORATORY

Electronic Voting Machine with Fingerprint Authentication

Presented by: Group 6

2006055 : Sadman Samin Rahul

2006056 : Sumaiya Tabassum Anha

2006057 : Mohammed Irtisam Sajin

2006058: Asmaul Husna Pushpita

Presented to:

Md. Ehsanul Karim

Lecturer, EEE, BUET

Akif Hamid

Adjunct Lecturer, EEE, BUET

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Overview

- Background
- Design & Methodology
- Slave & Master Devices
- Mapping to COs and POs
- Bill of Materials

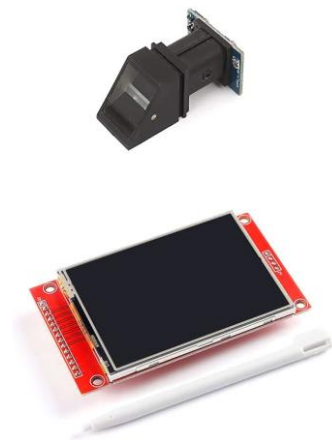
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Background

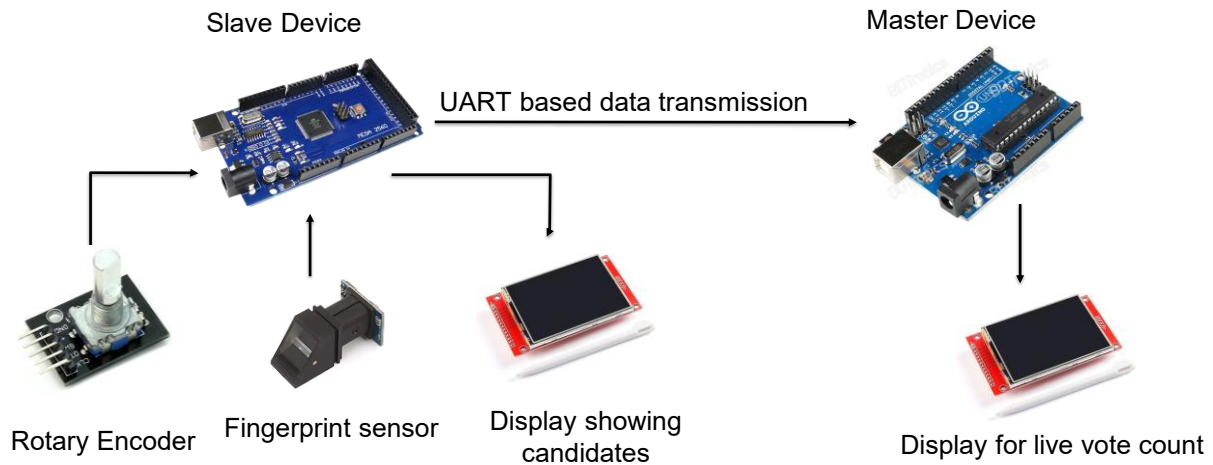
- Issues with traditional voting systems
- Manual errors & Impersonation
- Electronic Voting Machines (EVMs)
- Biometric verification.

Project Overview

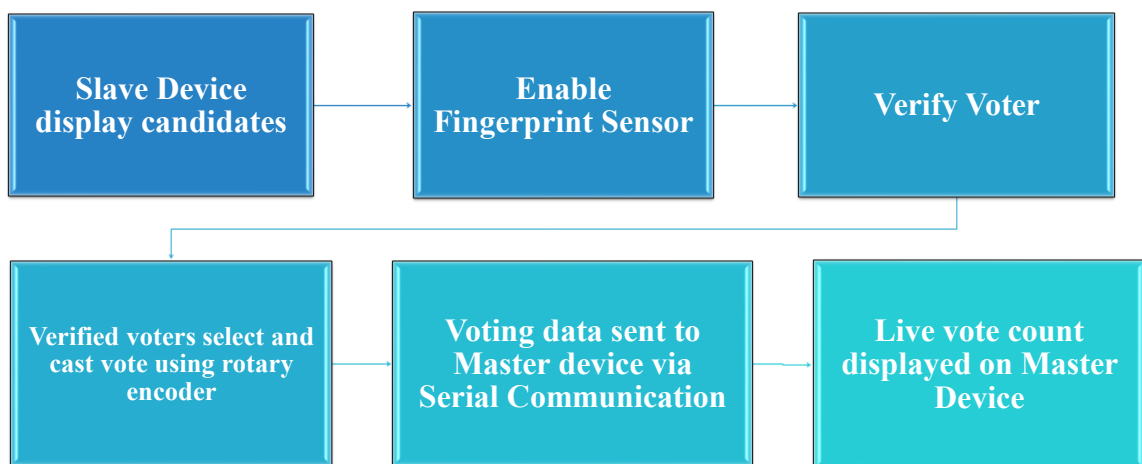
- Slave and Master devices
- UART communication
- Fingerprint verification
- Displays on both devices
- Rotary Encoder for smooth voting



Block Diagram



Project Methodology



Final Design

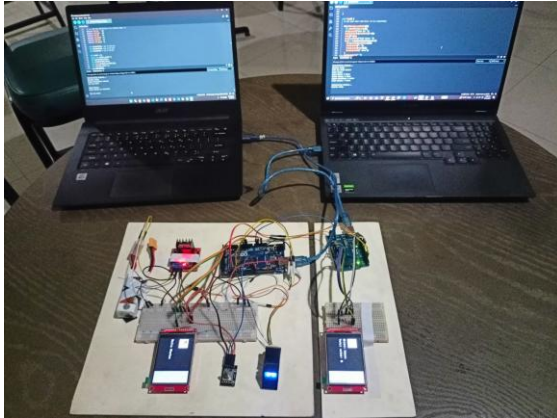
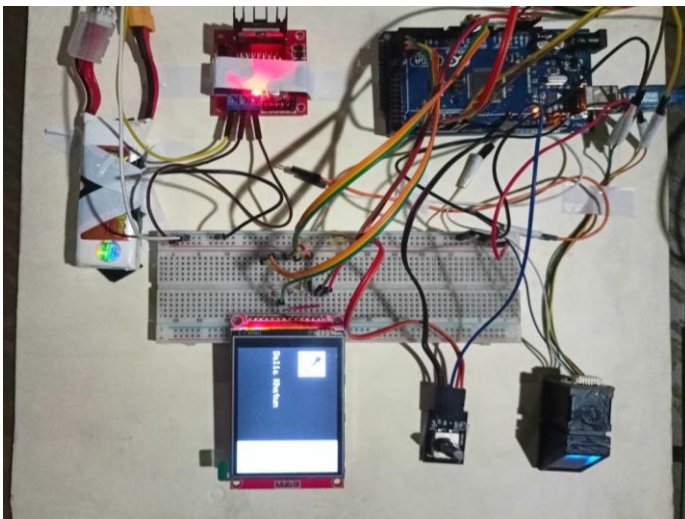


Figure: System Under Testing



Figure: Final Design – Slave Device(left) , Master Device(right)

Slave Device



```
You can vote!

Dalia Khatun
Chayan Chowdhury
Chayan Chowdhury is selected!

You can vote!

Dalia Khatun
Chayan Chowdhury
Shibli Noman
Shibli Noman is selected!

Already Voted!!!!

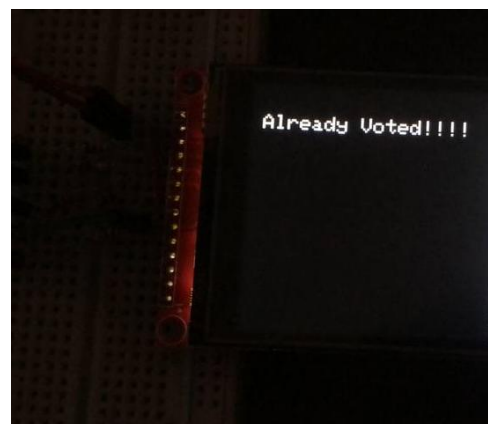
You can vote!

Dalia Khatun
Chayan Chowdhury
Shibli Noman
Shibli Noman is selected!
```

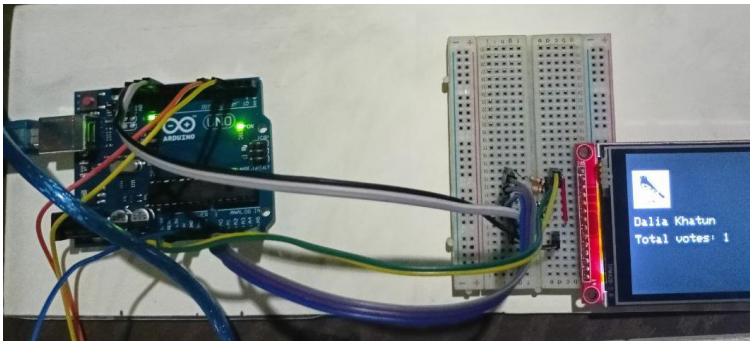
Slave Device Display



Slave Device Display



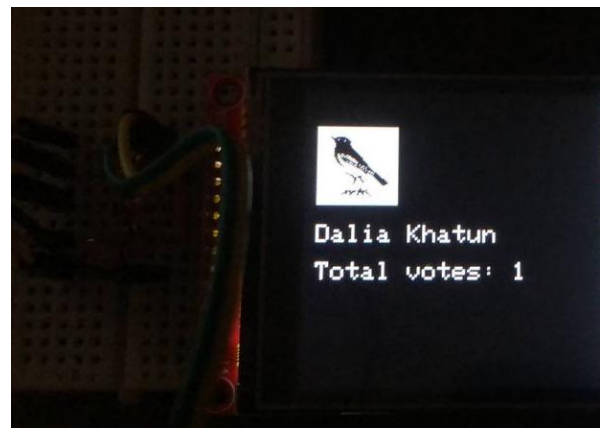
Master Device



```
Vote for Chayan Chowdhury
Total Votes:
Alice Ahmed: 0 votes
Shibli Noman: 0 votes
Chayan Chowdhury: 1 votes
Dalia Khatun: 0 votes
```

```
Vote for Dalia Khatun
Total Votes:
Alice Ahmed: 0 votes
Shibli Noman: 0 votes
Chayan Chowdhury: 1 votes
Dalia Khatun: 1 votes
```

Master Device Display



Mapping to COs and POs

CO	PO	How it is addressed
CO1	PO(e)	Utilize VerilogHDL, ARM assembly, C programming and microcontrollers boards for implementing and analyzing theoretical concepts at software and hardware levels
CO2	PO(c)	Design an embedded system with appropriate considerations to safety, cultural, societal, and environmental considerations
CO3	PO(f)	Assess impact of embedded systems project design on Societal, Health, Safety, Legal and Cultural Issues
CO4	PO(g)	Evaluate Sustainability the and Impact of the Designed embedded systems Project in the Societal and Environmental Contexts
CO5	PO(i)	Demonstrate membership and leadership in designing embedded system related problem solving
CO6	PO(j)	Communicate effectively on embedded system related design with presentation and detailed report
CO7	PO(k)	Demonstrate project management and cost analysis for embedded system project

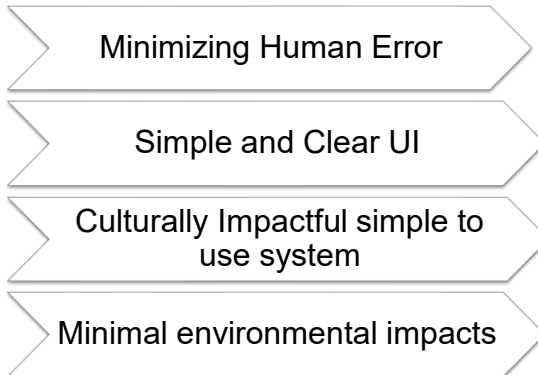
Mapping to COs and POs

CO1- PO(e) – Utilize VerilogHDL, ARM assembly, C programming and microcontrollers



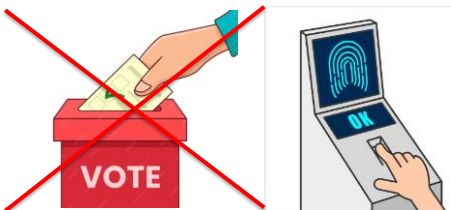
Mapping to COs and POs

CO2- PO(c) – Design an embedded system with safety, cultural, and environmental considerations



Mapping to COs and POs

CO3 – PO(f) – Assess impact on societal, health, safety, legal and cultural issues



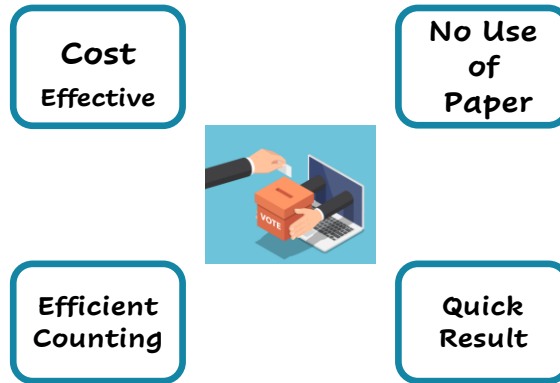
**No Use of Paper supports
eco-friendly practices**



**Preventing Conflicts
Between Groups**

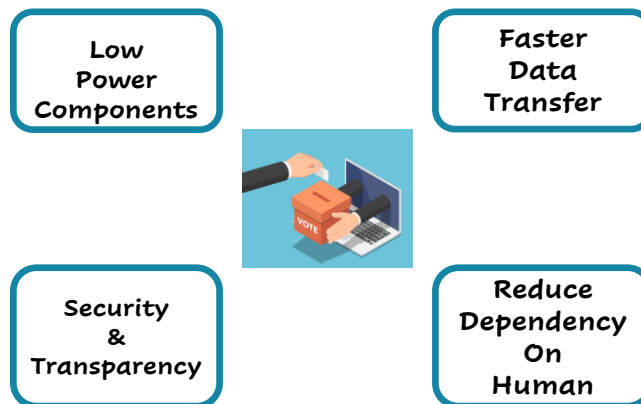
Mapping to COs and POs

CO4 – PO(g)– Impact in societal/environmental contexts



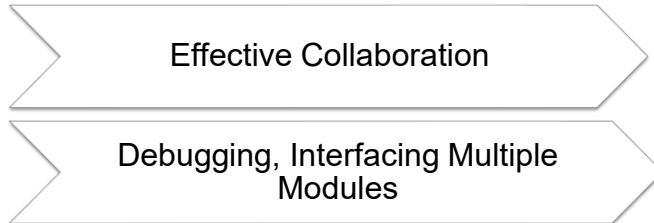
Mapping to COs and POs

CO4 – PO(g)– Evaluate sustainability



Mapping to COs and POs

CO5 – PO(i) – Demonstrate membership and leadership in solving embedded design problems



Mapping to COs and POs

CO7 – PO(k) – Demonstrate project management and cost analysis

The members contributed both in groups and as a whole to design and implement the overall project.

Individual Contributions:

- **2006055:** Display modules, Serial Communication
- **2006056:** Fingerprint Sensor, Algorithms & Logic Implementation
- **2006057:** Display Modules, Algorithms & Logic Implementation
- **2006058:** Fingerprint Sensor, Serial Communication

All the members collaborated for the final integration of the project.

Bill of Materials

Component	Quantity	Unit Price (Tk)	Total Price (Tk)
Arduino UNO	1	650	650
Arduino Mega	1	1800	1800
TFT 2.8" Display	2	1140	2280
R307 Fingerprint Sensor	1	1950	1950
Rotary Encoder	1	110	110
Battery (Lipo)	1	1350	1350
Resistors	6-10	-	15
Breadboard	2	145	290
Jumper Wires	1 set	100	100
PVC Boards	2	110	220
		Total=	8765

Thank you